National Institute of Mental Health Diagnostic Interview Schedule

Its History, Characteristics, and Validity

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• **A new interview schedule allows lay interviewers or clinicians to make psychiatric diagnoses according to *DSM-III* criteria, Feighner criteria, and Research Diagnostic Criteria. It is being used in a set of epidemiological studies sponsored by the National Institute of Mental Health Center for Epidemiological Studies. Its accuracy has been evaluated in a test-retest design comparing independent administrations by psychiatrists and lay interviewers to 216 subjects (inpatients, outpatients, ex-patients, and nonpatients).**

(*Arch Gen Psychiatry* 1981;38:381-389)

The National Institute of Mental Health (NIMH) Diagnostic Interview Schedule (DIS) is a highly structured interview designed to make diagnoses by the following three systems: (1) *DSM-III*, the criteria published by the American Psychiatric Association in 1980[1]; (2) the Feighner criteria, published in 1972, from the Washington University Department of Psychiatry in St Louis[2]; and (3) the Research Diagnostic Criteria (RDC), published by Spitzer et al in 1978.[3] It is possible to make diagnoses by all three systems with a single interview because the three systems share a common heritage. All reflect that common heritage in the degree to which they address diagnosis from a descriptive rather than etiological perspective. All assort symptoms into groups so that a positive diagnosis requires not only a minimum number of symptoms, but also a representative distribution of them; all require that symptoms meet a minimum standard of severity; and all require that symptoms not be explainable by physical illness or another psychiatric diagnosis. For certain diagnoses, each system also sets a maximum age at onset; and for others, each requires that the symptoms occur together and dissipate together, forming a discrete episode. Furthermore, all three systems agree on many of the specific symptoms of particular diagnoses. These similarities make it possible for many interview questions to serve the three systems at once.

See also pp 393, 400, 408, and 417.

The Feighner criteria and the RDC did not claim to provide diagnoses for all psychiatrically ill persons. Both have a residual of "undiagnosed" but psychiatrically ill cases that either fail to meet minimum criteria for the disorder they most resemble, have atypical ages for onset, satisfy criteria for multiple diagnoses with no clearly primary diagnosis, or show a symptom pattern that is not consistent with any of the syndromes contained in the system. The *DSM-III does* attempt to cover all psychiatrically ill persons. The DIS covers all diagnoses in the RDC and Feighner criteria and a selected group of *DSM-III* diagnoses (those that appear in the other two systems, plus diagnoses for which relatively detailed and specific criteria are provided). The diagnoses covered are given in Table 1.

HISTORY OF THE DIS

The DIS was written at the request of the Division of Biometry and Epidemiology of the NIMH for use in a set of large-scale epidemiological studies (the Epidemiological Catchment Area [ECA] projects). Our group at Washington University was invited to construct it along the lines of a recently completed instrument, the Renard Diagnostic Interview (RDI) (Helzer et al, p 393). The RDI was distinctive in that it fully specified all questions and probes to be used; its symptom scoring system indicated not only whether a symptom met criteria, but *why* it failed to; and it was accompanied by a set of computer programs that made diagnoses on the basis of analysis of symptom scores, thus minimizing the amount of clinical judgment required. At the time the NIMH was searching for an appropriate instrument, an assessment of the RDI's validity and reliability in psychiatric inpatients had just been completed. That assessment used a test-retest design in which lay interviewers were compared with each other, with psychiatrists using the RDI, and with psychiatrists using the Washington University Department of Psychiatry's traditional semistructured diagnostic interview. That test showed that lay interviewers (ie, interviewers without any previous training in psychology or psychiatry) after one week of training could obtain interviews that provided most Feighner diagnoses both reliably and validly.

Following an ad hoc advisory committee meeting of US diagnostic instrument experts chaired by the NIMH Biometry and Epidemiology Division Director, Darrel Regier, MD, the NIMH decided to support the design of a new

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Table 1.—Diagnoses Scored by DIS Computer Programs*

	DSM-III	Feighner (Renard)	RDC
290.xx	Senile and presenile dementias†	Organic brain syndrome	...
295.xx	Schizophrenic disorders	Schizophrenia†	Schizophrenia†
296.xx 296.3x 296.5x 296.6x	Major depression‡	Depression†	Major depressive disorder†‡
296.4x	Bipolar disorder, manic‡	Mania	Manic disorder
300.03	Panic disorder‡	Anxiety neurosis†	Panic disorder
300.21 300.22	Agoraphobia‡	Phobic neurosis	Phobic disorder
300.29	Simple phobia‡	...	...
300.30	Obsessive-compulsive disorder	Obsessive-compulsive neurosis	Obsessive compulsive disorder†‡
300.40	Dysthymic disorder	...	...
300.81	Somatization disorder§	Hysteria†§	Briquet's disorder§
301.70	Antisocial personality disorder†‡§	Antisocial personality disorder†§	Antisocial personality§
302.7x	Psychosexual dysfunction‡§	...	...
302.00	Ego-dystonic homosexuality§	Homosexuality§	...
302.5x	Transsexualism§	Transsexualism§	...
305.0x	Alcohol abuse	...	...
303.9x	Alcohol dependence	Alcoholism†	Alcoholism†
305.1x	Tobacco use disorder	...	...
305.2x	Cannabis abuse	...	...
304.3x	Cannabis dependence	Drug dependence	...
305.3x	Hallucinogen abuse	...	...
305.4x	Barbiturate abuse	...	...
304.1x	Barbiturate dependence	...	...
305.5x	Opioid abuse	...	...
304.0x	Opioid dependence	...	...
305.6x	Cocaine abuse	...	...
305.7x	Amphetamine abuse	...	...
304.4x	Amphetamine dependence	...	...
...	Other drug abuse	...	...
...	Other drug dependence	...	...
307.10	Anorexia nervosa	Anorexia nervosa	...
312.31	Pathological gambling‡	...	...

*DIS indicates the Diagnostic Interview Schedule; Feighner, Feighner criteria (Renard Diagnostic System); and RDC, Research Diagnostic Criteria.

†The diagnosis can be made at both "definite" and "probable" levels.

‡The diagnosis can be made with or without a preemptive diagnosis.

§Some or all of the symptoms appear in the DIS Supplement.

instrument that would permit the use of lay interviewers and have the general design and features of the RDI. In addition, it would incorporate a number of modifications so that it could provide current as well as lifetime diagnoses and diagnoses by *DSM-III* criteria and RDC, as well as by Feighner criteria. To make these modifications, the Washington University group was joined by Robert Spitzer, MD (who had led the design of the Schedule for Affective Disorders and Schizophrenia [SADS] interview[4] to make RDC diagnoses and who had been chairman of the task force to create *DSM-III*) and Janet B. W. Williams, MSW, (who had been coordinator of the *DSM-III* field trials).

CHARACTERISTICS OF THE DIS

Constructing an interview that allows a lay interviewer to perform the many tasks involved in making a diagnosis is a complex job. The multiple decisions that must be made can be illustrated with the diagnosis of panic disorder (or what was called "anxiety neurosis" under the Feighner system). Table 2 gives diagnostic criteria from the following three sources: *DSM-III*, Feighner criteria, and RDC. An analysis of these criteria shows that the following questions must be answered before a diagnosis can be made:

1. Did a panic attack occur? The answer is "yes" if (*a*) it was a discrete episode (ie, no symptoms before or after); (*b*) it was characterized by feelings of apprehension, fearfulness, or sense of doom; (*c*) it included a minimum of four characteristic symptoms out of 12 *(DSM-III)*, or four out of six (Feighner), or three out of ten (RDC); (*d*) it was sufficiently severe to lead to seeing a physician, taking medication more than once, or interfering significantly with a person's life (Feighner, RDC); and (*e*) it was *not* explainable by physical exertion, a life-threatening situation, or a physical disorder.

2. Have a sufficient number of attacks occurred? The answer is "yes" if there is a history of at least three *(DSM-III)* or six (Feighner, RDC) attacks.

3. Are the attacks repetitive rather than isolated? The answer is "yes" if three occurred in three weeks (*DSM-III* only).

4. Are the attacks characteristic of the person's life rather than confined to a brief, atypical period? The answer is "yes" if attacks occurred in six different weeks (Feighner) or over a period of six weeks (RDC).

5. Did attacks begin before a maximum age? The answer is "yes" if the attacks occurred before age 40 years (Feighner).

6. Were the attacks explainable as symptoms of another psychiatric disorder? The answer is "no" if the attacks were not part of a major depressive disorder, somatization disorder, or schizophrenia *(DSM-III);* if the first attack preceded symptoms of any other disorder by at least two years (Feighner); or if the attacks did not occur within two months of an episode of depression, schizophrenia, or schizoaffective disorder (RDC).

7. Is the person tense, nervous, or high-strung between attacks? The answer must be "yes" according to Feighner and RDC; this is common but not required, according to *DSM-III*.

The Feighner and RDC diagnostic systems have been characterized as belonging to the "Chinese menu system of diagnosis," implying that diagnosis is chiefly a matter of counting symptoms that have been sorted into groups. Our display of the requirements for a diagnosis of panic disorder shows that detection of the presence of symptoms is only one part of the diagnostic process. Symptom counts

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Table 2.—Diagnostic Decision Elements

DSM-III, Panic Disorder	Feighner Criteria, Anxiety Neurosis	Research Diagnostic Criteria, Panic Disorder
	Frequent and "Abnormal" Attacks	
At least 3 panic attacks within 3-wk period in circumstances other than during marked physical exertion or in life-threatening situation. Not due to physical disorder	At least 6 anxiety attacks each separated by at least 1 wk from others. They must occur at times other than marked physical exertion or life-threatening situations and in absence of medical illness that *could* account for symptoms of anxiety	At least 6 anxiety attacks distributed over 6 wk for definite diagnosis; 3 attacks over 3 wk for probable diagnosis
	Required Symptoms	
Panic attacks are manifested by discrete periods of apprehension or fear	Anxiety attacks are manifested by apprehension, fearfulness, or sense of impending doom	Panic attacks do *not* last most of day. Discrete periods of apprehension or fearfulness
At least 4 of following symptoms during each attack: Dyspnea Palpitations Chest pain or discomfort Choking or smothering sensation Dizziness, vertigo, unsteady feelings Paresthesias Feelings of unreality Hot and cold flashes Sweating Faintness Trembling or shaking Fear of dying, going crazy, or doing something uncontrolled	At least 4 of following symptoms are present during majority of attacks (or 2 for probable diagnosis): Dyspnea Palpitations Chest pain or discomfort Choking or smothering sensation Dizziness Paresthesias	At least 3 of following symptoms during majority of attacks for definite diagnosis if current (2 if past); 2 symptoms for probable diagnosis if current (1 if past): Dyspnea Palpitations Chest pain or discomfort Choking or smothering sensation Dizziness, vertigo, or feelings of unreality Paresthesias Sweating Faintness Trembling or shaking Fear of dying, going crazy, or doing something uncontrolled
No "nervousness" criterion, but common between attacks	Chronic nervousness	Nervous between attacks
	Severity	
No severity criteria specified; "clinically significant"	Severity criteria required	Severity criteria required
	Exclusion Criteria	
Not due to another mental disorder such as major depression, somatization disorder, or schizophrenia	In presence of other psychiatric illness or illnesses, this diagnosis is made only if criteria antedate onset of other psychiatric illness by at least 2 yr	Do not coincide with or predate or postdate by less than 2 mo major depression, schizophrenia, or schizoaffective disorders
Not only precipitated by circumscribed phobic stimulus	No phobic criterion	Not if limited to circumscribed phobic stimulus
	Age	
No age criterion, but usually late adolescence or early adult life, and *may* begin in midadult life	Age at onset prior to 40 yr	No age criterion

apply to criteria 1b and 1c. All the remaining criteria concern severity, frequency, distribution over time, and eliminating possible alternative explanations for the presence of the symptoms.

The DIS undertakes to make these complex decisions for all the diagnoses that appear in Table 1. It will be noted in Table 1 that five diagnoses require use of a supplement. This supplement contains questions that were not cleared for use in the ECA projects by the US Office of Management and Budget (OMB). (Clearance from the OMB is required for all population survey research done under a US government contract or cooperative agreement.) In addition to the diagnoses given in Table 1, the DIS provides a summary of the specific drug abuse and drug dependence diagnoses in *DSM-III* and adds specific drugs of dependence to the Feighner criteria. It also provides a diagnosis of "none of the ascertained diagnoses" for each system in lieu of a diagnosis of "well." For Feighner criteria and RDC, which discriminate between "probable" and "definite" diagnoses, it provides this discrimination wherever these diagnostic systems do. For *DSM-III*, which discriminates between substance abuse and dependence, the interview makes *that* distinction. Where *DSM-III* and RDC call for preemptive diagnoses, the DIS provides diagnoses both with and without the preemptions. Thus, it is possible to discern whether positive criteria for each diagnosis are met even when that diagnosis is excluded because of the presence of another disorder.

All diagnoses are made on a lifetime basis first, and then the interview inquires as to how recently the last symptom was experienced. On the basis of these answers it decides whether the disorder is current, defined for four time periods: the last two weeks, the last month, the last six months, and the last year. It also ascertains the age at the last symptom; the age at which the first symptom appeared; and, in the upcoming version 3, it will ascertain whether medical care was ever sought for symptoms of that disorder.

In addition to diagnostic results, the DIS also provides a total symptom count across diagnoses for each of the three systems and a count of the number of criteria met for each diagnosis, whether or not that diagnosis is positive.

To allow making diagnoses by computer, the interview covers each criterion in the form of one or more precoded, closed-ended questions. Questions in the main body of the interview ascertain the presence of symptoms and whether they meet criteria for frequency and clustering in time and whether they occurred early enough to meet the age-

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at-onset criterion. Figure 1 shows the interview section used to evaluate these criteria for panic disorder. To decide whether symptoms meet criteria for severity and whether they are explained by physical illness, injury, alcohol, or drugs, an adjunct to the interview, the Probe Flow Chart, is used (Fig 2). If any one of four questions in the Probe Flow Chart is answered positively (telling a doctor about the symptom, telling any other professional, taking medication for it more than once, having it interfere with one's life significantly), the symptom is judged clinically significant. If none of these criteria is positive, the symptom is coded 2, or "not clinically significant." If a doctor said the symptom was due to a psychiatric disorder or "nerves" or if it ever occurred when it was not caused by physical illness, alcohol, medicine, or drugs, it is considered a psychiatric symptom. If the symptom was always due to alcohol, or drugs, or to physical illness, it is coded accordingly and the cause is recorded for later review.

The Probe Flow Chart allows probes needed to make decisions about severity and alternative explanations to be as fully spelled out as the main question. Thus, virtually all questions can be read by lay interviewers as written. Rules about which probing questions to use for a particular symptom are also fully explicit and are indicated by the coding alternatives next to each question. (In Fig 1, the presence of numbers "12345" next to the coding blank for question 62 means that the full set of probes must be used for that question. The fact that only numbers 1 and 5 appear next to coding blanks for the 12 symptoms characterizing an attack means that a simple "yes" or "no" suffices for these questions.) The amount of discretion that the interviewer is to exercise, either in wording questions or in deciding when to probe, has been reduced to a minimum. This is what makes it possible to teach lay interviewers within one or two weeks to give an interview that provides the complex, clinical-like data necessary for diagnosis. After the interviewer follows these explicit instructions, the actual diagnosis is made by the computer.

While the DIS goes a long way toward regulating the interaction between respondent and interviewer, there remain interviewing skills to be taught. These include not only how to develop rapport with a respondent, but also how to use nondirective probes to clarify ambiguous answers and how to keep respondents' attention on answering the questions to avoid inappropriate responses and to control the time required to complete the interview. The interviewer must understand each question in order to judge whether a satisfactory answer has been obtained. In the hands of a well-trained lay interviewer or clinician, the DIS takes between 45 and 75 minutes for most respondents.

The DIS resembles the instruments developed 20 years ago to do surveys in the general population, instruments like the Home Interview Survey used in the Midtown Manhattan Study[5] and the Health Opinion Survey used in the Stirling County Study,[6] in that it requires the interviewer only to score answers to specified questions, not to improvise questions or make psychiatric judgments. Indeed, it goes further in this direction than those instruments by specifying probes as well. But it differs substantially from these earlier instruments in that it covers a large number of diagnoses comprehensively, including all necessary criteria, while they only sampled symptoms of a restricted number of diagnoses. (That sample of symptoms was much more representative of the diagnoses of anxiety disorder, depression, and obsessive-compulsive disorder

DIS VERSION 3:

QUESTIONS RELATING TO PANIC DISORDER (ANXIETY NEUROSIS)

61. Have you ever considered yourself a nervous person? 1 5 ____

62. Have you ever had a spell or attack when all of a sudden you felt frightened, anxious, or very uneasy in situations when most people would not be afraid? 12345 ____

IF TOLD DOCTOR (Q.62), CODE 5.
OTHERS CODE 1. ____

IF Q.62 IS CODED 2-5, ASK A-E.
OTHERS SKIP TO Q.63.

A. During one of your worst spells of suddenly feeling frightened or anxious or uneasy, did you ever notice that you had any of the following problems?

ASK AND CODE EACH SYMPTOM, ADDING "DURING THIS SPELL" BEFORE EACH.

During this spell--

(1) were you short of breath--having trouble catching your breath? . . . . . 1 5 ____

(2) did your heart pound? . . . . . 1 5 ____

(3) were you dizzy or light-headed? . . . . . 1 5 ____

(4) did your fingers or face tingle? . . . . . 1 5 ____

(5) did you have tightness or pain in your chest? . . . . . 1 5 ____

(6) did you feel like you were choking or smothering? . . . . . 1 5 ____

(7) did you feel faint? . . . . . 1 5 ____

(8) did you sweat? . . . . . 1 5 ____

(9) did you tremble or shake? . . . . . 1 5 ____

(10) did you feel hot or cold flashes? . . . . . 1 5 ____

(11) did things around you seem unreal? . . . . . 1 5 ____

(12) were you afraid either that you might die or that you might act in a crazy way? . . . . . 1 5 ____

B. How old were you the first time you had one of these sudden spells of feeling frightened or anxious? AGE: ____ ____

WHOLE LIFE = 02
IF DON'T KNOW AND AGE UNDER 40, CODE 01.
IF DON'T KNOW AND AGE 40 OR MORE, ASK: "Would you say it was before or after you were 40?"

BEFORE 40 = 01
40 OR MORE = 88
DON'T KNOW = 99

C. Have you ever had 3 spells like this close together--say within a 3-week period? 1 5 ____

D. Have spells like this occurred during at least 6 different weeks of your life? 1 5 ____

E. Have you had a spell like this within the last two weeks?

IF YES, CODE 1.
IF NO, ASK: When did you last have a spell like this? ____

WITHIN LAST 2 WEEKS 1
WITHIN LAST MONTH 2
WITHIN LAST 6 MONTHS 3
WITHIN LAST YEAR 4
MORE THAN 1 YEAR AGO 5

IF MORE THAN A YEAR:

How old were you then? AGE: ____ ____

Fig 1.—Section of Diagnostic Interview Schedule (DIS) version 3 used to evaluate criteria for panic disorder (anxiety neurosis). Note that questions are precoded and closed ended.

than of other diagnoses.) The DIS makes diagnoses based on well-defined criteria, while these earlier studies made no diagnoses at all, but rather depended on a clinician to review a summary of the interview protocol to decide how closely the respondent resembled a psychiatric case. By using clear diagnostic algorithms for its computer programs, the DIS allows no irrelevant data to enter into the diagnostic process. In the earlier studies, the clinician had access to interview information about the childhood home, demographics, treatment experience, physical illnesses, and attitudes, as well as information from hospital and social agency records. These data may have biased his

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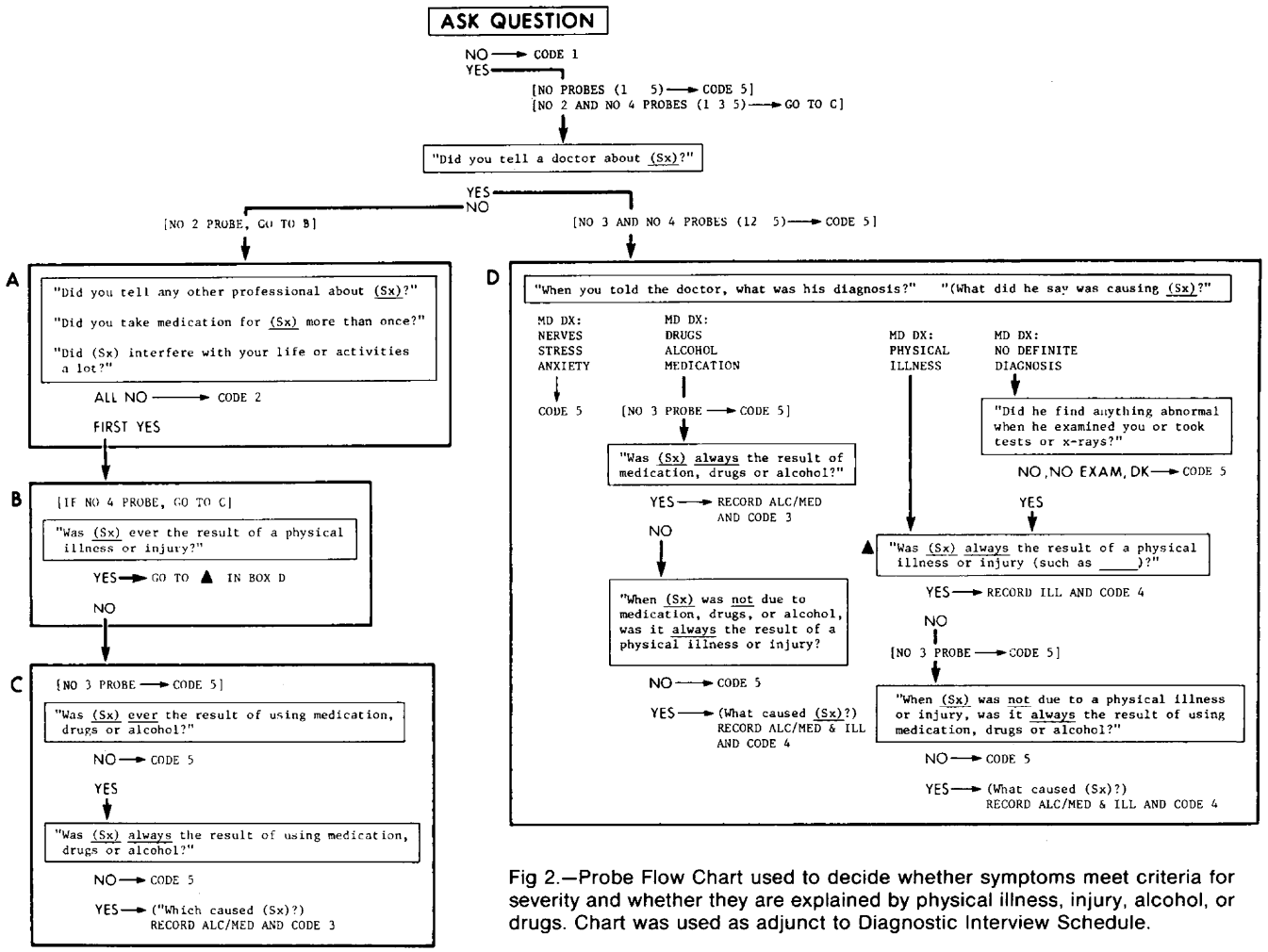


Fig 2.—Probe Flow Chart used to decide whether symptoms meet criteria for severity and whether they are explained by physical illness, injury, alcohol, or drugs. Chart was used as adjunct to Diagnostic Interview Schedule.

judgment as to whether or not he was looking at a "case." The computer, unlike the clinician, looks only at those items to which the program directs it in making a diagnostic decision.

More recently, research diagnoses in clinical populations (and to some extent in general populations) have been made using the SADS[4] and the Present State Examination (PSE).[7] The DIS resembles the SADS and the PSE in making specific diagnoses and in bringing to bear on the diagnostic process not only the presence or absence of symptoms, but judgments about their severity, their clustering in time, their frequency, and possible alternative explanations for their occurrence. It differs from both in covering many more diagnoses, in covering diagnoses according to *DSM-III* criteria (which will presumably be the standard for clinical practice in the United States), and in avoiding screening questions and "cutoffs." When cutoffs are used in the SADS or PSE, information about criteria for a diagnosis is not obtained if either another diagnosis that preempts it has been found positive or if a few criterion symptom questions have been answered in the negative. Not only may some positive diagnoses be missed, but the total number of symptoms is then not available, since different numbers of symptoms will have been asked of different subjects. When diagnoses in these two systems are represented only by screening questions, diagnoses cannot be made without seeking external information from hospital records or informants.

The DIS does provide optional "skipouts" that can be used after enough negative or positive answers have been obtained so that no matter how subsequent questions were answered, the diagnosis would remain the same. These shorten the interview without the risk of underdiagnosis associated with screening questions and cutoffs in SADS and PSE. But using the optional skipouts does distort the count of total symptoms and may change ages at onset or remission, and therefore their use is not recommended.

The DIS also differs from the PSE in providing lifetime as well as current diagnoses and differs from both the PSE and SADS in defining "current" flexibly so that results can be compared with the one-month definition of the PSE or the one-year definition of the SADS. But the DIS's most important difference from both the PSE and the SADS lies in the degree to which questions and probes have been specified (so that the interviewer is not called on to improvise questions) and the degree to which scoring of symptoms follows automatically from answers to the questions posed (to minimize the clinical judgment required). These features make possible the use of lay interviewers after a fairly brief training period.

Furthermore, the DIS has the advantage of serving more than one diagnostic schema, so that the alternative systems can be compared. It also offers flexibility with respect to using or ignoring diagnostic hierarchies, which makes it possible to compare different methods for establishing hierarchies.

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Table 3.—Adequacy of Lay Interviewers in Giving DIS Version 2: Specific Lifetime Diagnoses Without Exclusion Criteria*

Diagnosis	Diagnostic System	No. of Positive Cases			%	
		Psychiatrist	Lay Interviewer	K	Sensitivity	Specificity
Anorexia nervosa	*DSM-III*	9	9	1.00	100	100
	Feighner	10	10	1.00	100	100
Pathological gambling	*DSM-III*	13	12	.96	92	100
Alcohol						
Dependence/definite or abuse/probable	*DSM-III*	77	68	.86	86	98
	Feighner	68	68	.85	90	95
	RDC	73	73	.85	90	95
Dependence/definite	*DSM-III*	52	45†	.80	79	97
	Feighner	51	50	.80	84	95
	RDC	63	57†	.88	87	99
Organic brain syndrome						
Mild, severe	. . .	20	22	.79	85	97
Severe	. . .	11	11	.90	91	99
Tobacco use disorder	*DSM-III*	108	106	.78	89	90
Drug						
Dependence/definite or abuse/probable	*DSM-III*	26	34†	.73	88	94
	Feighner	33	39	.76	88	94
Dependence/definite	*DSM-III*	23	27	.77	87	96
	Feighner	26	30	.88	96	97
Phobias	*DSM-III*					
	Agoraphobia	53	55	.67	77	91
	Simple	39	41	.47	59	88
	Feighner	63	62	.53	67	86
	RDC	67	71	.58	75	85
Mania	*DSM-III*	31	26	.65	65	97
	Feighner	29	25	.66	66	97
	RDC	31	26	.65	65	97
Antisocial personality	*DSM-III*	16	20	.63	75	96
	Feighner					
	Definite, probable	40	44	.73	83	93
	Definite	25	30	.69	80	94
	RDC					
	Definite, probable	12	12	.56	58	97
	Definite	9	7	.61	56	99

TEST OF THE VALIDITY OF THE DIS

Method

While the interview was specially designed to enable lay interviewers to obtain psychiatric diagnoses comparable with those a psychiatrist would obtain, its actual ability to achieve that goal needed careful evaluation. To evaluate the instrument, we conducted interviews with 216 persons, each of whom were interviewed twice (once by a lay interviewer and once by a psychiatrist, both using the DIS). Lay interviewers were hired for this study through the medical school personnel office. None had had previous experience in interviewing, and none had had any clinical training. All were college graduates. The psychiatrists had recently completed their training. Three had been trained at our medical school; one was trained elsewhere.

In addition to giving the DIS, the psychiatrists scored their clinical impressions with respect to each diagnosis covered after completion of the DIS. They then had a free question period during which they could ask any further questions they wished, either to get new information or to clarify answers obtained during the DIS interview. At the conclusion of the free answer period, they again scored their clinical impressions.

Differences in computer diagnoses between a psychiatrist's and lay interviewer's DIS interview with the same subject indicate to what extent a lay interviewer can give the DIS as well as a psychiatrist can. Questions asked during the free period indicate areas in which the psychiatrist thought the DIS was incomplete or unsatisfactory. Differences between clinical impressions before and after the free question period indicate areas of the DIS that are actually deficient (that is, where the clinician's uncertainties could be resolved through more extensive questioning). When a diagnosis remains uncertain after the free question period, it can be assumed that the difficulty lay in the subject's ability to answer or in the complexities of the clinical picture, rather than in the instrument.

The order in which the two interviews were given was randomized. At the final count, there were 97 in which the lay interviewer was the first interviewer and 119 in which the psychiatrist was the first interviewer.

Subjects included 118 psychiatric inpatients, 39 psychiatric outpatients, 24 enrollees with the Washington University Medical Care Group with no known psychiatric disorder, ten members of Gamblers Anonymous, and 26 ex-patients. One staff member was assigned to look at patients' charts to determine what diagnoses were likely to be found and to invite the participation of those whose diagnoses met our needs. The interviewers were not informed as to the suspected diagnosis and were warned not to say anything about the interview itself until the second interview had been completed. All interviews were tape-recorded, and one or two for each interviewer were videotaped.

Interviews were edited for completeness, with different editors for lay interviewers and psychiatrists to assure that there would be no bias toward concordance in the editing process. After editing, the codes were entered into the computer and diagnostic programs applied. Quotas for each diagnosis were filled on the basis of the computer diagnosis of the psychiatrist's interview.

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Table 3.—Adequacy of Lay Interviewers in Giving DIS Version 2: Specific Lifetime Diagnoses Without Exclusion Criteria (cont)

Diagnosis	Diagnostic System	No. of Positive Cases: Psychiatrist	No. of Positive Cases: Lay Interviewer	κ	%: Sensitivity	%: Specificity
Depression	*DSM-III*	99	93	.63	80	84
	Feighner					
	Probable, definite	87	83	.66	78	86
	Definite	73	65	.63	71	90
	RDC					
	Probable, definite	99	96	.64	80	84
	Definite	86	84	.68	80	87
Obsessive-compulsive disorder	*DSM-III*	30	28	.60	63	95
	Feighner	28	24	.60	61	96
	RDC	30	26	.60	63	95
Schizophrenia	*DSM-III*	34	32	.60	65	94
	Feighner					
	Probable, definite	26	20	.51	50	96
	Definite	13	11	.56	54	98
	RDC					
	Probable, definite	40	42	.54	65	90
	Definite	23	28	.57	70	93
No covered *DSM-III* diagnosis	*DSM-III*	24	23	.59	63	96
Psychosexual dysfunction	*DSM-III*	71	72	.56	72	84
Somatization disorder	*DSM-III*	32	15†	.50	41	99
	Feighner					
	Definite, probable	29	21	.59	55	97
	Definite	22	16	.65	59	98
	RDC	30	16†	.52	43	98
Panic disorder	*DSM-III*	39	29†	.40	44	93
	Feighner					
	Definite, probable	37	32	.56	59	94
	Definite	24	20	.59	58	97
	RDC					
	Definite, probable	42	32	.51	52	94
	Definite	40	29†	.53	53	95

*Feighner indicates Feighner criteria (Renard Diagnostic System); RDC, Research Diagnostic Criteria. Two hundred sixteen subjects were interviewed twice for organic brain syndrome and "no covered *DSM-III*" diagnoses; 204 subjects were interviewed twice for other diagnoses (excludes 12 whom one or both interviewers judged too organically disordered to be asked symptom questions for other diagnoses.

†McNemar's test ($\chi^2 > 3.84$) indicates that the result is biased.

Quotas required a minimum of ten positive cases for each diagnosis, and no more than 20 were sought for any particular diagnosis. We were able to fill the minimum quota for all disorders except dysthymia, for which we had nine cases. The number of positive cases per diagnosis varied from nine to 116 for psychiatrists' interviews (Table 3) and between seven and 118 for lay interviewers' interviews (Table 3). Approximately half the cases had tobacco use disorder or depression; one third qualified for psychosexual dysfunction or alcohol dependence or abuse; one fourth had phobias. These five diagnoses are so commom among psychiatric patients that quotas for them were oversubscribed. Organic brain syndrome quotas were filled with some difficulty because appropriate patients were often too demented to give informed consent or too physically ill to participate.

Results

The study findings are still under analysis. The following four measures of lay interviewers' ability to provide correct psychiatric diagnoses can be presented at this time: (1) bias in lay interviewers' diagnoses; (2) κ as a measure of concordance between lay interviewers' and psychiatrists' diagnoses; (3) sensitivity (percentage of cases correctly identified) and specificity (percentage of noncases correctly identified) of the lay interviewers' diagnoses, using the psychiatrists' interviews as the yardstick; and (4) the lay interviewers' credibility (ie, the proportion of false-positives when a lay interviewer makes a positive diagnosis). In these analyses,

Table 4.—Validity of DIS Version 2: Means Across Lifetime Diagnoses Without Exclusion Criteria*

	No. of Diagnoses Made	κ	%: Sensitivity	%: Specificity
All diagnoses				
DSM-III	20	.69	75	94
Feighner	20	.70	74	95
RDC	14	.62	67	93
Diagnoses present in both systems				
DSM-III and Feighner	14†	.67	72	94
	13	.71	77	95
DSM-III and RDC	11†	.63	67	94
	10	.65	68	94
Feighner and RDC	14	.64	68	94
	14	.62	67	93

*The means are weighted by the number of positive cases. DIS indicates the Diagnostic Interview Schedule; Feighner, Feighner criteria; and RDC, Research Diagnostic Criteria.

†*DSM-III* appears to make one more diagnosis than the other systems because phobias are divided into agoraphobia and simple phobias in *DSM-III*.

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Table 5.—Lay Interviewer Credibility*

Diagnosis	(No. Positive) % of Cases Confirmed by Psychiatrist: *DSM-III*	Feighner	RDC
Pathological gambling	(12) 100	...	...
Anorexia nervosa	(9) 100	(10) 100	...
Alcohol Dependence	(45) 91	(50) 86	(57) 96
Abuse, dependence	(68) 97	(68) 90	(73) 90
Tobacco use disorder	(106) 91	...	...
Organic brain syndrome Severe	(11) 91	...	...
Mild, severe	(22) 77	...	...
Somatization disorder Definite	(15) 87	(16) 81	(16) 81
Probable, definite	...	(21) 76	...
Depression Definite	93 (82)	(65) 80	(84) 82
Probable, definite	(102) 76†	(83) 82	(96) 82
Mania	(26) 77	(25) 76	(26) 77
Phobias	...	(62) 68	(71) 70
Agoraphobia	(55) 75	...	...
Simple	(41) 56	...	...
Drug Dependence	(27) 74	(30) 83	...
Abuse, dependence	(34) 68	(39) 74	...
Psychosexual dysfunction	(72) 71	...	...
Schizophrenia Definite	(32) 69	(11) 64	(28) 57
Probable, definite	...	(20) 65	(42) 62
Obsessive-compulsive disorder	(28) 68	(24) 71	(26) 74
No covered *DSM-III* diagnosis	(23) 65	...	...
Antisocial personality Definite	(20) 60	(30) 67	(7) 71
Probable, definite	...	(44) 75	(12) 58
Panic disorder Definite	(29) 59	(20) 70	(29) 72
Probable, definite	...	(32) 69	(32) 69
Mean (weighted by No. positive)	(...) 74	(...) 77	(...) 74

*Credibility refers to how often a positive diagnosis (without exclusion criteria) was confirmed by a psychiatrist's interview. RDC indicates Research Diagnostic Criteria; Feighner, Feighner criteria (Renard Diagnostic System).

†The diagnoses here are depression or dysthymia.

Table 6.—Subjective Clinical Impressions After DIS*

Lifetime Diagnosis	% of Cases: Psychiatrist in Doubt About Diagnosis After Completing DIS	Doubts That Could Be Resolved by Further Questions
Antisocial personality	7	3
Depression	4	2
Somatization disorder	4	2
Alcohol dependence, abuse	3	2
Mania	3	2
Dysthymia	2	2
Anorexia nervosa	2	2
Phobias	4	1
Panic disorder	3	1
Obsessive-compulsive disorder	1	1
Schizophrenia	3	0.5
Drug dependence, abuse	2	0.5
Organic brain syndrome	1	0.5
Psychosexual dysfunction	1	0
Pathological gambling	0	...

*DIS indicates the Diagnostic Interview Schedule. N = 204.

we will use diagnoses *without* hierarchies because we want to be able to discuss differences in ascertaining criteria for a particular diagnosis without the confusion of differences in making some other diagnosis that may account for the presence or absence of exclusion criteria.

We explored whether the lay interviewers produced biased results through the McNemar test, which asks whether discrepant cases tend to be significantly more positive or negative for lay interviewers than for psychiatrists. Of the 52 diagnostic comparisons made, seven were significantly biased. Lay interviewers tended to underdiagnose alcohol dependence, somatization disorder, and panic disorder by *DSM-III* and RDC (but not by Feighner criteria) and to overdiagnose *DSM-III* drug abuse (but not *DSM-III* drug dependence or probable or definite drug dependence by Feighner criteria).

The greatest bias involved somatization disorder. Somatization disorder is the disorder that most often requires a decision as to whether a symptom is due to medical rather than psychiatric causes. We had asked lay interviewers to accept medical explanations offered by respondents without attempting to judge their plausibility. If we ignore medical explanations and count both medically explained and not medically explained symptoms as positive, the bias disappears. But the psychiatrist is then overdiagnosing somatization disorder by 30%. To prevent bias without overdiagnosing when the interview is given by a lay interviewer, a physician should review the reported "medical" explanations in those interviews where medical explanations prevent the diagnosis of somatization disorder to judge whether or not such medical explanations are plausible.

Table 3 gives the κ value, sensitivity, and specificity for each of the diagnoses in each diagnostic system. Ego-dystonic homosexuality and transsexualism are omitted because the appropriate questions had not been cleared by NIMH for submission to OMB. The diagnoses of antisocial personality, psychosexual disorder, and somatization are included because all symptom questions had been approved for this validity study, even though some were not cleared for the ECA project. Results are based on 216 twice-interviewed subjects for organic brain syndrome and "no covered *DSM-III* diagnosis" and on 204 twice-interviewed subjects for other diagnoses. The 204 excludes 12 cases in which one or both interviewers judged the subject to be too demented to give answers to questions that would have permitted making other diagnoses. All κ values are greater than .40, and all but one (*DSM-III* panic disorder) are greater than or equal to .50. Sensitivity ranges from 100% to 41%; specificity ranges from 100% to 84%. It is interesting that diagnoses in which symptoms are determined by questions considered "sensitive," and therefore most likely to be denied or refused, show the best results. Gambling, alcoholism, drug abuse, and antisocial personality were among the most accurately made diagnoses, even though making those diagnoses required asking about illegal activities and problem drinking.

Table 4 gives means for κ values, sensitivity, and specificity across diagnoses. For *DSM-III* diagnoses, the mean κ was .69, the mean sensitivity was 75%, and the mean specificity was 94%.

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Table 7.—Validity of DIS in Former vs Current Patients*

Lifetime Diagnosis	κ	(No.) % of Cases Sensitivity	Specificity
Depressive disorder			
Current patients	.59	(75) 79	(67) 81
Former patients	.49	(17) 82	(9) 67
Alcohol abuse or dependence			
Current patients	.87	(59) 88	(83) 98
Former patients	.92	(11) 91	(15) 100
Tobacco use disorder			
Current patients	.79	(78) 88	(64) 91
Former patients	.92	(16) 100	(10) 90
Psychosexual disorder			
Current patients	.57	(54) 78	(88) 81
Former patients	.57	(10) 60	(16) 93

*The diagnoses are *DSM-III* lifetime diagnoses with at least ten positive cases according to psychiatrists' interviews. Current patients who were considered too demented to answer questions about their diagnosis were excluded.

These means were calculated both as simple averages of the 20 diagnoses obtained and as weighted averages, weighting each value by the number of positive cases in the psychiatrists' interviews. Results were identical by the two methods. Results for Feighner diagnoses were similar to those for *DSM-III*, while results for RDC diagnoses were less good. However, the lower RDC scores apparently were due only to the fact that the diagnoses made by RDC happened to be concentrated among disorders with lower concordance. When comparisons were restricted to disorders that appeared in both systems, results were comparable among the three diagnostic systems.

The mean κ values reported here are excellent compared with those reported in the literature for the reliability of other instruments. However, it is difficult to make comparisons between this study and others because methods varied among them. So far as we know, this is the only study to have included such a variety of sources of subjects or to have set quotas for specific disorders. Since measures of reliability and validity vary with base rates in the sample, these methodological variations probably have an effect on the results.

Table 5 gives lay interviewers' credibility, defined as the proportion of lay interviewers' positive diagnoses that were corroborated by the psychiatrists' diagnoses. The average was three out of four in each of the three systems, with a range from 100% for gambling and anorexia nervosa to a low of 56% for *DSM-III* simple phobia. While credibility scores have mean values similar to sensitivity scores, they lack values as low as those found for sensitivity.

To judge how satisfactory the answers provided through the DIS were to the psychiatrists interviewing when they were called on to decide whether a diagnosis was present or absent by their best clinical judgment, we examined how often they were left with questions after completing the DIS. Table 6 shows that they rarely were uncertain about the diagnoses that the DIS covered. It also shows that the opportunity to ask further questions resolved their doubts in less than half the cases. Thus, only half their uncertainties could be attributed to deficiencies in the DIS.

Another concern we addressed was whether the DIS would be effective with persons who did not have a current complaint. Many of our subjects were inpatients who had recently had to communicate their psychiatric problems to a physician. This might have made it easier for them to answer the questions posed by the DIS. To learn whether recency of illness mattered, we compared the concordance, sensitivity, and specificity of interviews with current patients vs ex-patients who had not been hospitalized for at least three years. Table 7 gives the four diagnoses for which at least ten positive cases were found in the psychiatrists' interviews with ex-patients. It will be noted that there is no particular pattern of advantage according to current or past patient status. For depressive disorder, κ values are higher for current patients; for alcoholism and tobacco use disorder, κ values are higher for former patients; and for psychosexual disorder there is no difference. Sensitivity and specificity are both noticeably greater for ex-patients in one diagnosis and for current patients in another and are approximately equal in the two other diagnoses.

COMMENT

Finally, we need to ask just what we are studying with the test-retest design we have described. Is it the reliability of the DIS, or its validity? Ideally, to study the validity of an instrument, one would like to have an absolute standard to compare it against. In this study, we have appointed the DIS in the hands of a psychiatrist as a yardstick against which we are measuring its performance in the hands of a lay interviewer. We would have preferred to have used an independent instrument as that yardstick, but the difficulty was in finding another measure that covered the full range of *DSM-III* diagnoses made by the DIS. To our knowledge, no previous instrument has been designed to make *DSM-III* diagnoses, and those that make the same diagnoses by other criteria often themselves have not been tested for validity by some external standard. Our use of repeated interviews with the same instrument has led some to say that we are studying reliability, not validity. However, our study does not follow the classic reliability model either. In studying reliability by test-retest methods, one treats the two applications of the instrument equally. The κ values we have presented do this, but our use of specificity and sensitivity shows that we are assuming that the psychiatrist's interview is the correct one against which the lay interviewer's work is to be judged. In short, like many prior efforts to establish validity, our work is a bootstrapping operation. We are using an imperfect yardstick against which to measure the lay interviewer. We hope that the good results we have obtained are in spite of these difficulties and not because of them.

A third version of the DIS has been prepared, with modifications stemming from what we learned during our validity study. This version will undergo extensive testing during the ECA projects, and those tests may result in further improvements. While the DIS will undergo further development, at present it appears to be a reasonably satisfactory instrument for making psychiatric diagnoses, whether in the hands of clinicians or in the hands of lay interviewers.

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